

Underwater Acoustics

A Brief Introduction

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Decibel

Gain of a system is usually expressed as the logarithmic ratio of the strength of the output signal to the strength of the input signal. Like all ratios, this form of gain is unitless. However, there is an actual unit intended to represent gain, and it is called the *bel* [7].

As a unit, the bel was actually devised as a convenient way to represent power *loss* in telephone system wiring rather than *gain* in amplifiers. The unit's name is derived from Alexander Graham Bell, the famous American inventor whose work was instrumental in developing telephone systems. Originally, the bel represented the amount of signal power loss due to resistance over a standard length of electrical cable. It was later decided that the bel was too large of a unit to be used directly, and so it became customary to apply the metric prefix *deci* (meaning 1/10) to it, making it *decibels*, or dB. Now, the expression "dB" is so common that many people do not realize it is a combination of "deci-" and "-bel," or that there even is such a unit as the "bel" [7].

If we want to express the power gain of a signal with respect to reference power P_r , then

$$P(\text{dB}) = 10 \log_{10}(P/P_r)$$

Since power is proportional to the square of voltage, we can write this as

$$P(\text{dB}) = 20 \log_{10}(V/V_r)$$

In some cases, we may want to express signal amplitude in decibels instead of signal strength. Then, our reference will be volts instead of watts, and

$$V(\text{dB}) = 20 \log_{10}(V/V_r) = 20 (\log_{10}(V) - \log_{10}(V_r))$$

We will try to clarify these definitions in the following.

Understanding the Transducer and Hydrophone Specs

Transducer and hydrophone specifications usually include the Open Circuit Receiving Response (OCRR), Transmitting Voltage Response (TVR), and the directivity pattern. In the following, we will define these properties and use the ITC1001 spherical transducer specifications as an example.

Open Circuit Receiving Response (OCRR) is defined as the output voltage (V) generated by the transducer per μPa of sound pressure as a function of frequency. OCRR is expressed in dB re 1V/ μPa . The OCRR for the ITC1001 is given in Figure 1. At $f_c=22$ kHz, the OCRR value is -190dB re 1V/ μPa . If the received sound intensity level (SIL) at the transducer is 190 dB re μPa , then at the output of the transducer we will measure

$$V_{\text{dB}} = \text{SIL} + \text{OCRR}(f_c) = 190 + (-190) = 0 \text{ dB},$$

Since V_{dB} is relative to 1 V, we can write it as

$V_{dB} = 10 \log_{10}(V/1)$. Then the output voltage in Volts is

$$V = 10^{(V_{dB}/10)} = 1 \text{ V.}$$

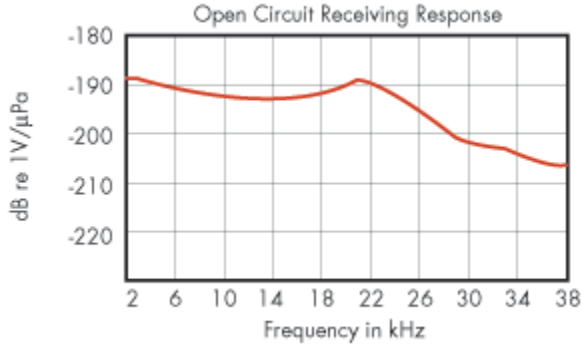


Figure 1 OCRR for ITC1001 spherical transducer.

Transmitting Voltage Response (TVR) is defined as the output sound intensity level (SIL) generated at 1m range by the transducer per 1 V of input Voltage as a function of frequency. The TVR for ITC1001 is given in Figure 2. At $f_c=22$ kHz, the TVR value is 144dB re $\mu\text{Pa} / 1\text{V} @ 1\text{m}$. If the input voltage is 200 V, the sound intensity level (SIL) at 1m range will be

$$\text{SIL} = \text{TVR}(f_c) + V_{dB}$$

Since V_{dB} is relative to 1 V, $V_{dB} = 10 \log_{10}(V/1)$. Then

$$\text{SIL} = 144 + 10\log_{10}(200) = 144 + 26 = 170 \text{ dB re } \mu\text{Pa}$$

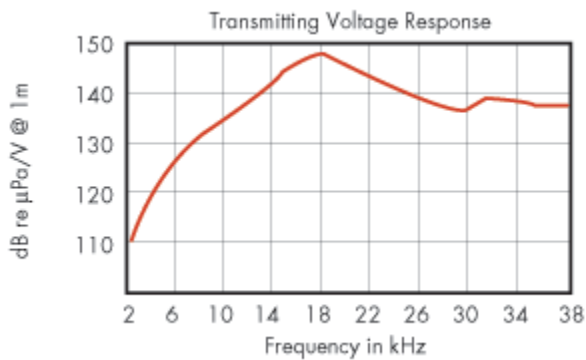


Figure 2 TVR for ITV1001 spherical transducer.

Directionality Pattern is defined as the SIL as a function of angle on horizontal and vertical planes at a given frequency. The directivity pattern of the ITC1001 transducer for the horizontal plane is given in Figure 3. This transducer has same directivity pattern in the vertical plane also; hence it is a spherical transducer.

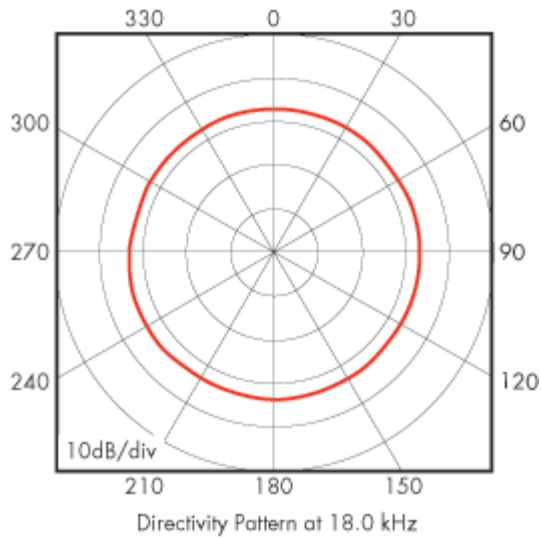


Figure 3 Directivity pattern of ITC1001 spherical transducer.

Toroidal transducers usually present a directivity pattern on the vertical plane similar to the one given in Figure 4. Their directivity on the horizontal plane is similar to that of Figure 3, therefore resulting in a toroid in 3D.

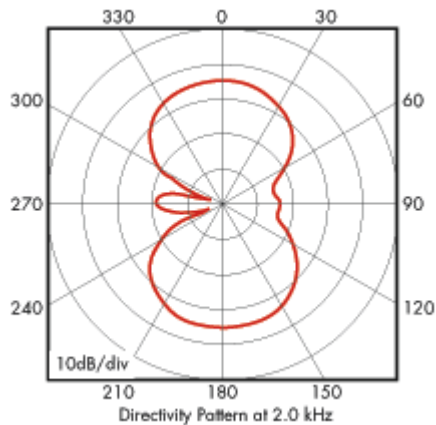


Figure 4 Directivity pattern of ITC2010 toroidal transducer on the vertical plane. 0 degree represents the horizontal direction.

Hydrophone Pre-Amplifiers

The signal level at the output of a hydrophone is usually small, in the order of millivolts (mV). Hydrophones located in deep water are connected to a surface station through a cable of several 100 m, which introduces a loss in the signal strength. This loss may become so large that we may lose the signal and observe just noise. Therefore, we amplify the output of the hydrophone before sending it through the cable. The amplifier used for this purpose is called the pre-amplifier and is usually located right after the hydrophone under the water. The main purpose of the pre-amplifier is to amplify the signals so that they can travel over long cables until they reach the processing unit, which

is usually at the surface. At the processing unit, if needed, the signals are amplified one more time. If the cable is short, we may not need additional amplification.

The pre-amplifier voltage gain is listed in dB. That is,

$$\begin{aligned} V_{\text{gain}} &= 10 \log_{10}(V_{\text{in}}/V_{\text{out}}) \\ &= 10 \log_{10}(V_{\text{in}}) - 10 \log_{10}(V_{\text{out}}) \end{aligned}$$

For example, if we apply a signal of 10mV to a 23 dB gain pre-amplifier, we can calculate the output voltage as

$$\begin{aligned} V_{\text{out}}(\text{dB}) &= V_{\text{in}}(\text{dB}) + 23 \\ V_{\text{in}}(\text{dB}) &= 10 \log (V_{\text{in}}) = 10 \log (10\text{e-}3) = -20 \\ V_{\text{out}}(\text{dB}) &= -20 + 23 = 3 \text{ dB} \\ V_{\text{out}} &= 10^{(3/10)} = 1.9953 \text{ Volt } (\approx 2) \end{aligned}$$

Acoustic Channel Estimation

The first step in designing a communication system is to determine the channel characteristics. Once we determine the important parameters of the channel, we can design our signals to best fit the channel and optimize the system performance. In this section, we will review methods to estimate some important channel parameters.

Shallow Water:

In a shallow water channel, the acoustic waves travel through a direct path and also by bouncing from the surface and bottom. We can roughly estimate the propagation of acoustic signals over a shallow water channel by simplifying our environment parameters. If we assume that

- the sound speed is almost constant,
- surface and bottom are smooth,

we can geometrically calculate the expected propagation paths for the acoustic waves.

Once we determine the propagation paths (which is referred as ray tracing), we can estimate the received signal and power given the transmitted signal and source and destination locations. This type of propagation, where there are multiple rays that reach a receiver, is called multipath propagation.

Let's first consider the following shallow water channel, which we will refer as SWCH-1 (see Figure 5). The water depth is 100m and the distance between the source and receiver is 100m. For simplicity, we assumed that both source and the receiver are at 20m. The acoustic waves will reach to the receiver through several paths: the direct path (yellow), one bounce (green and red), one surface and one bottom reflection (magenta), and so forth. At each reflection, the acoustic waves will experience a loss in power in addition to the propagation losses. If we send a pulse of duration T ms and amplitude A V, we can calculate the received signal through ray tracing.

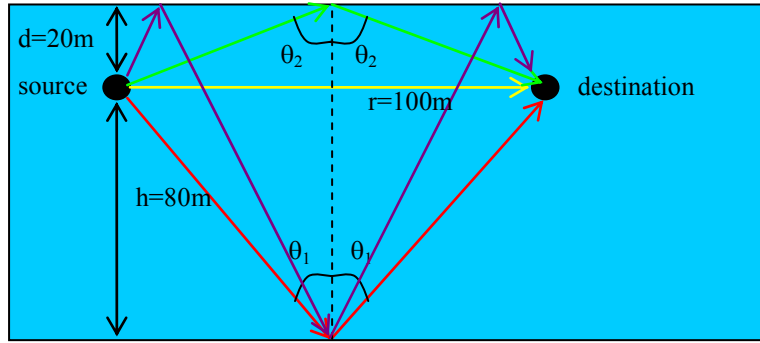


Figure 5 Short range shallow water channel (SWCH-1).

- 1) Calculate the length of propagation paths, for the direct path, one reflection paths, two reflection paths, and three reflection paths.
- 2) For each path, calculate the time of arrival to the receiver assuming a uniform sound speed of $c=1500$ m/s.
- 3) For each path, calculate the transmission loss, which is the sum of all reflection losses and the propagation loss. Assume that the center frequency, f_c , of the pulses is 22 kHz, sea water temperature, T , 15°C, and reflection loss of 1 dB at the surface and 3dB at the bottom.
- 4) Assume that we use an ITC1001 transducer as our transmitter and drive it with a 400 Vrms source. Determine the SIL at 1 m.
- 5) Determine the received SIL at the receiver for each path.
- 6) Determine the output voltage of the receiver for each path assuming that the OCRR of the hydrophone at 22 kHz is -162 dB re 1V/ μ Pa, and we employ a preamplifier with gain 40 dB.
- 7) Repeat these calculations for a range of 1000 m

Determining the Range of a Source

If a target transmits pulses, we can determine the range of that target by measuring the received pulses. The pulse is a sinusoidal wave of duration T_s , which can be written as

$$p(t) = A \sin(2\pi f_c t), 0 < t < T_s$$

and zero elsewhere.

We can write the received signal as a delayed and diminished version of the transmitted pulse as

$$r(t) = B \sin(2\pi f_c (t - \tau)), \tau < t < (T_s + \tau).$$

If we correlate the received signal with the transmitted signal, we can obtain an estimate of the time delay. Let's first define correlation.

Correlation:

Correlation is used to determine the similarity between two signals, $a(t)$ and $b(t)$, at various delay values, and is defined as

$$R_{(a,b)}(\lambda) = \int_{-\infty}^{\infty} a(t)b(t-\lambda)dt$$

The bigger the correlation value, the more similar the two signals are. If $b(t) = a(t)$, then the correlation becomes

$$R_{(a,a)}(\lambda) = \int_{-\infty}^{\infty} a(t)a(t-\lambda)dt$$

and is called auto-correlation of a . The auto-correlation will be maximum when the delay, λ , is zero.

Let's assume that $b(t)$ is the delayed version of $a(t)$, that is $b(t) = a(t-\tau)$. The correlation, $R_{(a,b)}$, will reach its maximum at $\lambda=\tau$. We can use this property to determine the delay value, τ .

- 1) Based on the path propagation delays you calculated in the previous section, what should be the maximum pulse width, T_s , for ranges of 100 and 1000 m. If the pulse duration is longer than T_s , what will happen?
- 2) If you know the transmitted pulse and time when the pulse was transmitted, how can you estimate the propagation delay of the direct path based on the received signal?
- 3) A transponder is a device that listens for a pulse at a frequency, f_1 , and responds with a pulse at a frequency, f_2 , after a delay of τ_t . If the target has a transponder, how can you determine the range of the target?

Determining the Direction of the Target

If the two hydrophones are separated by d meters, we can measure the propagation time at each hydrophone. If the propagation time measured at hydrophone A is less than the hydrophone B, we can say that the target is closer to hydrophone A. If our propagation time estimates are accurate, we can even calculate the angle of the target.

As a simple solution, we can simplify the problem to determine if the target is on the left or right half plane. Then, by comparing the propagation time to hydrophone A, which is located on the left side of the surface craft, r_A , and the propagation time to hydrophone B, which is located on the right side of the surface craft, r_B , we can determine the location of the target. In other words, if $(r_A > r_B)$, then the target is on the right half plane, else on the left half plane.

If we assume that the range of the target is much bigger than the separation between the two hydrophones, we can approximate the sound waves as plane waves, as shown in Figure 6. (r_1-r_2) is the difference between the propagation lengths of the waves, which is related to the propagation time difference.

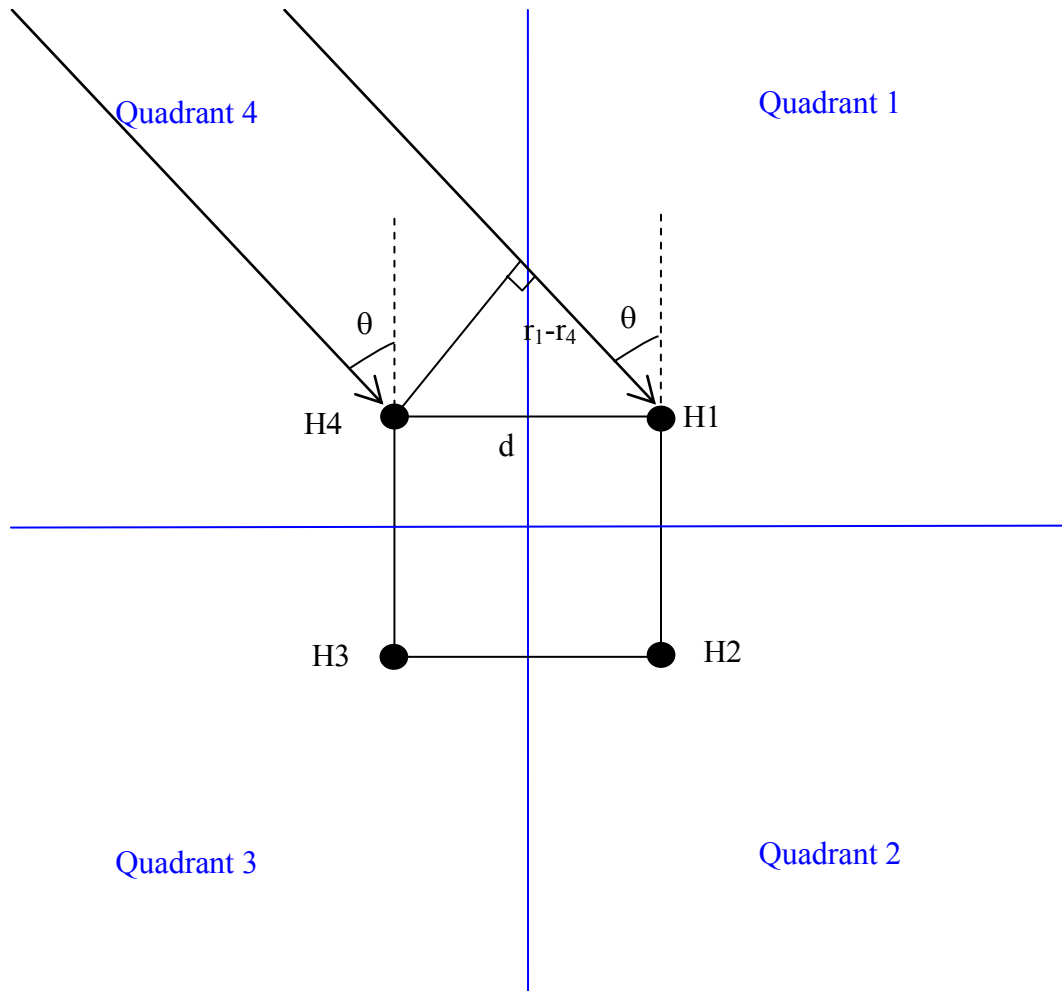


Figure 6 The surface craft has four hydrophones at each corner, separated by d meters. The target is far enough that the sound wave arrive to the surface craft as plane waves. The angle of incidence is θ .

How can you determine in which quadrant the target is located using all four hydrophones?

Underwater Acoustic Propagation Modeling Software

For modeling the acoustic propagation over underwater channels we will use a software suit called the Acoustic Toolbox [1], written by Mike Porter. The original software was written in Fortran, which is not a commonly used programming language anymore. Alec Duncan provided a Matlab front-end [2] for the Acoustic Toolbox. In this section, we will provide an introduction to the Acoustic Toolbox and the user friendly Matlab front-end.

The Acoustic Toolbox provides estimates for acoustic propagation of signals through the underwater channel by numerically solving propagation equations. The toolbox implements the following underwater acoustic propagation models:

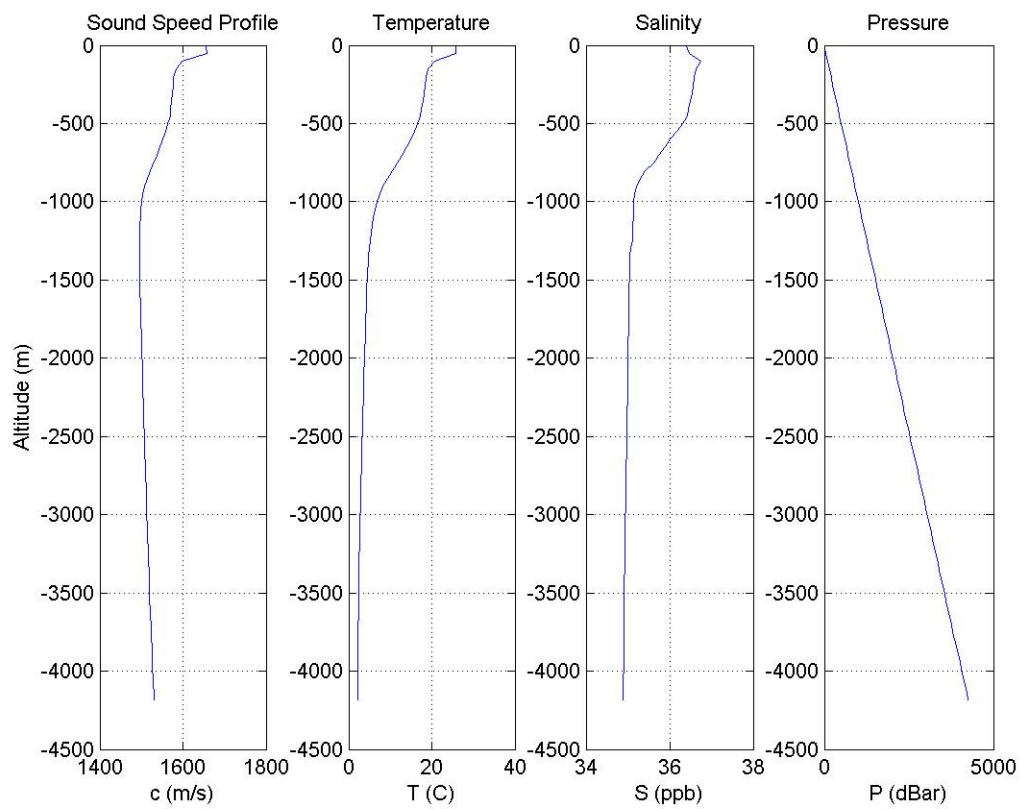
- Kraken normal mode model [3], [4]: a normal mode code for range-varying environments in either Cartesian (line sources) or cylindrical (point sources) coordinates
- KrakenC complex normal mode model [5]: a complex normal mode code for range-varying environments in either Cartesian (line sources) or cylindrical (point sources) coordinates
- Scooter fast-field model: a finite element code for computing acoustic fields in range-independent environments based on direct computation of the spectral integral with pressure and material properties approximated by piecewise-linear elements
- Bellhop ray and Gaussian beam tracing model [6]: a program which computes acoustic fields in oceanic environments via beam tracing, with the environment being an acoustic medium with a sound speed which can depend on range and depth

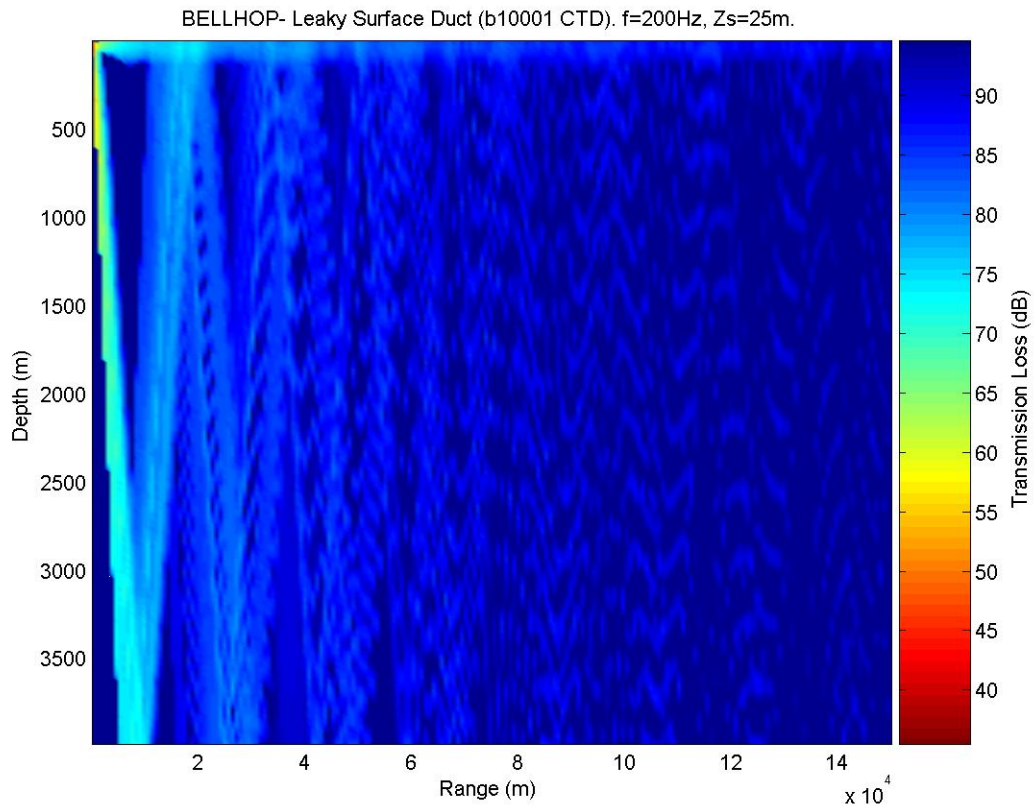
In addition, the toolbox can calculate Bounce bottom reflection coefficients for layered media.

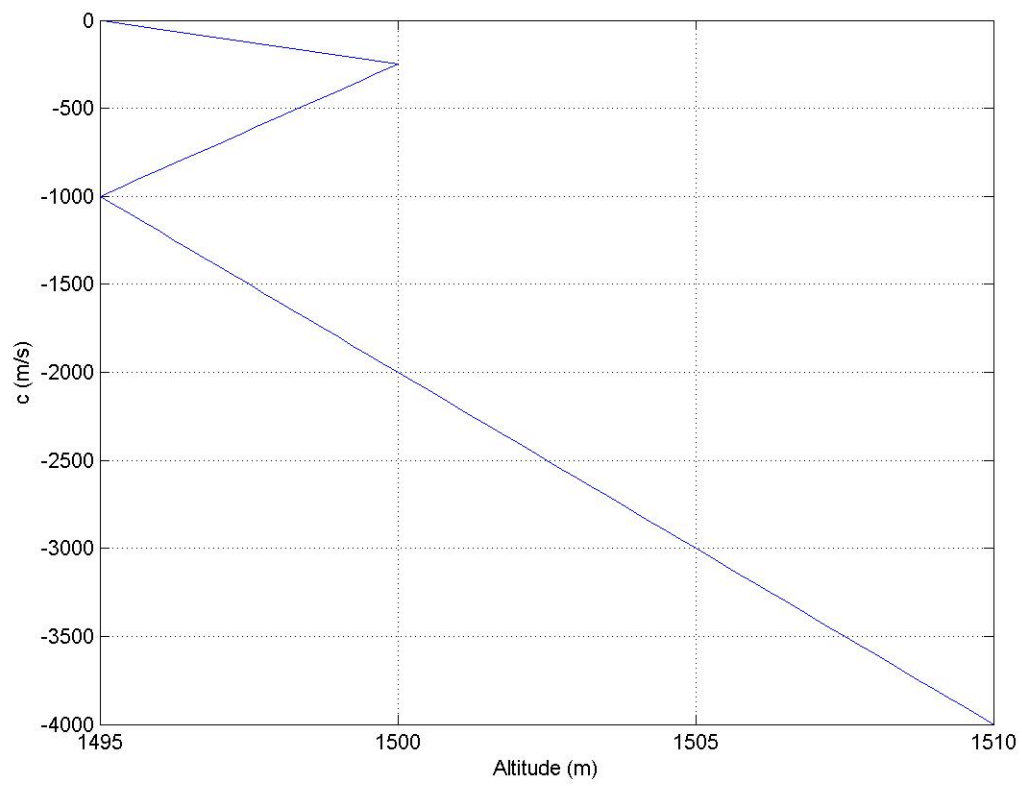
Acoustics Toolbox Front-End Users Manual

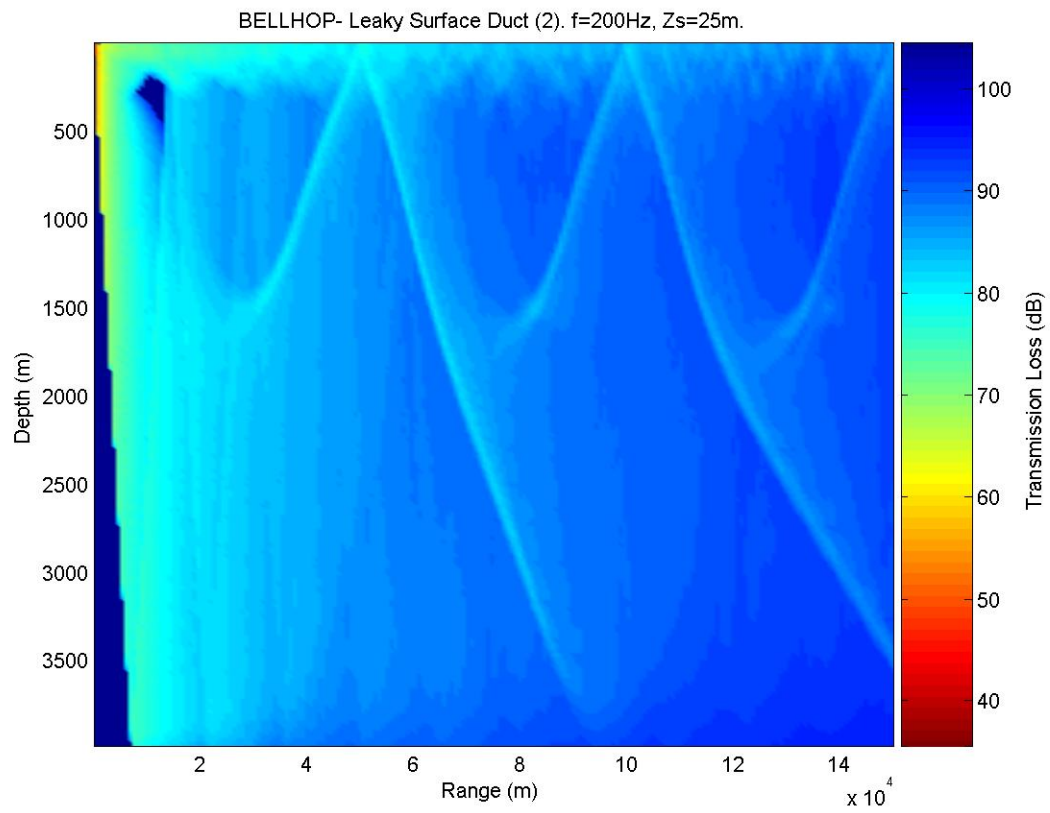
The first step for the calculation of transmission loss and ray trace, you will need to define your environment. In this course, we will only focus on the effect of the sound speed profile and bottom reflections on the acoustic propagation. We will use the default bottom definition provided by the program.

We will estimate the sound speed profile using the depth, temperature, and salinity information (measured by a CTD device). You can find real CTD data on some web sites or make your own measurements. For our example channel, we used the data set provided on the Bermuda Atlantic Time-Series Study web site [7], which can also be accessed through an ftp site [8]. We used a data set recorded in November 1988.









References

- [1] Acoustic Toolbox, URL:
<http://stommel.tamu.edu/~baum/linuxlist/linuxlist/node7.html#AcousticsToolbox>
- [2] Matlab front-end for Acoustic Toolbox, URL:
<http://www.curtin.edu.au/curtin/centre/cmst/products/actoolbox/>
- [3] M.B. Porter and E.L. Reiss, "A numerical method for ocean-acoustic normal modes," *J. Acoust. Soc. Am.*, Vol. 76, pp. 244-252, July 1984
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- [6] M.B. Porter and H.P. Buckner, "Gaussian beam tracing for computing ocean acoustic fields," *J. Acoust. Soc. Amer.*, Vol. 82, pp. 1349-1359, 1987.
- [7] http://www.allaboutcircuits.com/vol_3/chpt_1/5.html